APP2: Candy Crush

Algorithmic Problem Solving

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Learning objectives:

- Understand the limitations of greedy algorithms.
- Use dynamic programming to find an optimal solution to a problem.
- Identify the cases where dynamic programming can be used.

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I. APP2: Candy Crush

I.1. APP Objectives

After this APP2, you will be capable of:

- Knowing the principles of the following various types of problem-solving techniques: greedy algorithms, complete solution space exploration and dynamic programming;
- Evaluating the optimality and complexity of these methods on problems;
- Identifying the cases where dynamic programming is applicable and find the optimal solution.

I.2. Session Organization

Follow the guidelines of the previous project.

I.3. The Problem

Situation — problem

You want to open a candy shop and sell a new type of candy called BerLim (short for Berry and Lime). BerLim is a traditional stick-like candy, which has one-centimetre-long segments. Some segments taste like sweet berry, and others are sour, like lime. The manufacturer delivers BerLim in long sticks, which need to be split into several pieces at the border of some segments.

The children around you have sweet tooth, so they will only buy a piece of BerLim, if it contains more sweet segments than sour ones. Given the input described below, try to determine the maximum total length of BerLim that can be sold to the picky children, assuming that you break it up in the best possible way.

Input. The input begins with an integer M, the number of BerLim sticks that the manufacturer delivered. Then the details of each stick follows.

For each stick m ($1 \geq m \geq M$), the first line of input contains one integer N - the length of the stick in centimetres ($1 \leq N \leq 200$). The next line is a sequence of N characters '0' or '1', describing the segments of the stick from left to right ('0' denotes a sour lime segment, '1' - a sweet berry one).

Output. For each BerLim stick m, print out a line with a single integer: the total length of BerLim in centimeters that can be sold after breaking it up the best possible way.

Example.
Sample input:
2
15
100110001010001
16
0010111101100000

Sample output:
9
13

I.4. Resources

This APP will cover basic knowledge on greedy algorithms, solution space exploration and dynamic programming. It is required that you read the notes given at the address below. Moreover, if you feel you need more information, the book reference is extremely complete and give great details on this subject.

- Moodle - APS - Greedy Algorithms
- Moodle - APS - Dynamic Programming
- "Introduction to Algorithms" by Cormen, Leiserson, Rivest, and Stein (3rd edition);
  - Chapters 16 (Greedy algorithms);
  - Chapter 15 (Dynamic Programming);

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